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Digital mammography
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The Mammographic Image Analysis Society Digital Mammogram Database

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ABSTRACT

The Mammographic Image Analysis Society has produced a *Digital Mammography Database*. It contains 161 pairs of films and includes examples of abnormalities commonly encountered in screening as well as a comprehensive set of normal cases. The mammograms have been carefully selected from the United Kingdom National Breast Screening Programme to be of the highest quality of exposure and patient positioning. Each medio-lateral oblique view was digitised to a spatial resolution of 50μm pixel edge with a scanning microdensitometer with a linear response in the optical density range 0.0 to 3.2 with 8 bits representing each pixel. The entire database, when compressed, occupies less than 2 GBytes fitting onto a single 8mm magnetic tape. Copies are available for research purposes.

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INTRODUCTION

Computer vision systems which aim to assist, support or replace human operators must undergo stringent, comprehensive and appropriate acceptance testing. This is especially important in medical applications where, ultimately, computer vision may play a critical role in patient management. With the advent of Breast Screening Programmes in many western countries and the considerable research efforts currently being made in digital mammography, it is possible that the benefits of computer vision may soon be available to radiologists. Those who develop and test these algorithms have a responsibility to ensure their accuracy and robustness.

A common database is a positive step towards achieving consistency in performance comparison and the objective testing of algorithms. There are, however, many difficulties, both intellectual and practical with the establishment of such a library. Many of these issues were discussed by a group of panellists at the IEEE Biomedical Image Processing and Biomedical Visualisation Conference, San Jose, 1993. Reference to the proceedings [1] should be made for a comprehensive deliberation. In the United Kingdom the Mammographic Image Analysis Society (MIAS) considers the production of a mammographic database of importance, and agreed criteria for its design which recognised both financial and practical constraints:

- The number of digitised films (322) in the database was determined by time and financial restrictions.

- Mammograms should come from a *single* centre of the UK National Breast Screening Programme, reducing the organisational requirements of the database. Only the medio-lateral oblique (screening) view was available.

- The choice of digitiser was limited. The Joyce-Loebl scanning microdensitometer (sited at the Royal Marsden Hospital) was chosen, but this limits the grey-level resolution to 8 bits per pixel.

- The 50μm pixel edge was a compromise between scanning duration, image resolution and file size.

- The accumulation of supporting clinical information will necessarily take place over an extended period of time. The extent of the data may be limited due to retrospective collection.

These design criteria are not ideal. The database does however represent a constructive and practical contribution to computer vision research in mammography and it is hoped that it will encourage the production of more extensive collections of data.

PRODUCTION

Original mammograms were carefully selected from a major centre participating in the United Kingdom National Breast Screening Programme, ensuring high quality exposure and patient positioning. Bilateral medio-lateral oblique views were obtained for each case. Examples of both benign and malignant forms of abnormalities commonly encountered were included, as well as a comprehensive set of normals. Each breast type (fatty, fatty-glandular and dense) was represented in each category. Full details are given in table 1.

Digitisation was performed on a Joyce-Loebl scanning microdensitometer (SCANDIG-3) at 50μm square pixel edge and 8 bits per pixel. The device has a linear response in the range 0.0 to 3.2 optical densities [2]. Four image sizes were used to aid data management and reduce scanning times:

Small (**s**):	1600 pixels × 4320 pixels
Medium (**m**):	2048 pixels × 4320 pixels
Large (**l**):	2600 pixels × 4320 pixels
Extra Large (**x**):	5200 pixels × 4320 pixels

Filename mnemonics were generated by a prefix (**mdb**), a three digit serial number, **l** or **r** for left or right breast respectively and **s**, **m**, **l** or **x** to denote the image size; for example **mdb030rm**.

Each image was compressed with the UNIX *compress* command and stored on magneto-optical disk. The entire database of 322 films occupies less than 2GBytes and was written, in UNIX *tar* format, to a single 8mm magnetic tape. A *README* file was included with details of the database. Clinical information consists, initially, of the type of abnormality, if any, present on the mammogram. Additional clinical information is currently being collected for future distribution.

DISTRIBUTION

A copy of the database and single-site licence are available for a small fee which covers duplication costs. The information is made available for research purposes only can be obtained by contacting Email: MIASDB@ICR.AC.UK, or by contacting the correspondence author.

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Film Category	Breast Type			TOTALS	
	Fatty	Fatty-Glandular	Dense		
Calcifications	B: 2 M: 4	B: 5 M: 4	B: 5 M: 5	B: 12 M: 13	25
Circumscribed Masses	B: 7 M: 2	B: 6 M: 2	B: 3 M: 0	B: 16 M: 4	20
Spiculated Masses	B: 2 M: 4	B: 4 M: 3	B: 6 M: 2	B: 12 M: 9	21
Architectural Distortions	B: 4 M: 2	B: 2 M: 4	B: 4 M: 4	B: 10 M: 10	20
Asymmetries	B: 2 M: 3	B: 3 M: 2	B: 3 M: 4	B: 8 M: 9	17
Miscellaneous	B: 4 M: 4	B: 3 M: 2	B: 1 M: 1	B: 8 M: 7	15
Normals	66	64	74	204	204
TOTALS	106	104	112	322	322

Table 1: Summary of the contents of the MIAS digital mammography database (B-benign, M-malignant).

REFERENCES

1 Shtern F (moderator). Panel Discussion, Proc SPIE conference on Biomedical Image Processing and Biomedical Visualization 1993: 1905: 534-551.

2 Davies D H. Brit J Rad 1993: 66: 930-933.